



K. J. Somaiya College of Engineering, Mumbai-77
(A Constituent College of Somaiya Vidyavihar University)
Department of Computer Engineering

Batch: _____ **Roll No.:** _____

Experiment No. _____

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Implementation Matrix Chain Multiplication of Dynamic Programming

Objective: To learn Matrix chain multiplication using Dynamic Programming Approach

CO to be achieved:

CO 2 Describe various algorithm design strategies to solve different problems and analyse Complexity.

Books/ Journals/ Websites referred:

1. Ellis horowitz, Sarataj Sahni, S.Rajsekaran," Fundamentals of computer algorithm", University Press
2. T.H.Cormen ,C.E.Leiserson,R.L.Rivest and C.Stein," Introduction to algortihmts",2nd Edition ,MIT press/McGraw Hill,2001
3. <http://www.lsi.upc.edu/~mjserna/docencia/algofib/P07/dynprog.pdf>
4. <http://www.geeksforgeeks.org/travelling-salesman-problem-set-1/>
5. <http://www.mafy.lut.fi/study/DiscreteOpt/tspdp.pdf>
6. <https://class.coursera.org/algo2-2012-001/lecture/181>
7. <http://www.quora.com/Algorithms/How-do-I-solve-the-travelling-salesman-problem-using-Dynamic-programming>
8. www.cse.hcmut.edu.vn/~dtanh/download/Appendix_B_2.ppt
9. www.ms.unimelb.edu.au/~s620261/powerpoint/chapter9_4.ppt

Pre Lab/ Prior Concepts:

Data structures, Concepts of algorithm analysis

Historical Profile:

Dynamic Programming (DP) is used heavily in optimization problems (finding the maximum and the minimum of something). Applications range from financial models and operation research to biology and basic algorithm research. So the good news is that understanding DP is profitable. However, the bad news is that DP is not an algorithm or a data structure that you can memorize. It is a powerful algorithmic design technique.



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New Concepts to be learned:

Application of algorithmic design strategy to any problem, dynamic Programming method of problem solving Vs other methods of problem solving, optimality of the solution, Optimal Binary Search Tree Problems and their applications

Theory:

Problem definition:

Given a sequence of N matrices, the matrix chain multiplication problem is to find the most efficient way to multiply these matrices by minimizing the number of computations involved during multiplications.

Optimal Substructure: parameterization/ select the subgroup of matrices that will result in least number of computations.

For multiplication of matrix series A_i to A_j , choose A_k such that multiplication of matrices through $A_i \dots k$ and $A_{k+1} \dots j$ will incur least number of computations for any k such that $i \leq k \leq j$.

Recursive Formula:

$$n[i, j] = \begin{cases} 0 & i = j, \\ \min_{i \leq k < j} (m[i, k] + m[k + 1, j] + p_{i-1}p_kp_j) & i < j \end{cases}$$

Algorithm:



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Example:

Solution for the example:



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Analysis of algorithm:

CONCLUSION:



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Topic: Backtracking

Theory: In many applications of the backtrack method, the desired solution is expressible as an n -tuple $(x_1, \dots, x_n)$, where the x_i are chosen from some finite set S_i . Often the problem to be solved calls for finding one vector that maximizes (or minimizes or satisfies) a *criterion function* $P(x_1, \dots, x_n)$. Sometimes it seeks all vectors that satisfy P . For example, sorting the array of integers in $a[1 : n]$ is a problem whose solution is expressible by an n -tuple, where x_i is the index in a of the i th smallest element. The criterion function P is the inequality $a[x_i] \leq a[x_{i+1}]$ for $1 \leq i < n$. The set S_i is finite and includes the integers 1 through n . Though sorting is not usually one of the problems solved by backtracking, it is one example of a familiar problem whose solution can be formulated as an n -tuple.

Control abstraction:

```
void Backtrack( int k )
```

```
// This is a schema that describes the backtracking process //using recursion. On entering, the  
//first k-1 values x[1], x[2], //..., x[k-1] of the solution vector x[1:n] have been //assigned. x[]  
//and n are global.
```

```
{  
    for (each x[k] such that x[k] ∈ T(x[1], ..., x[k-1])  
    {  
        if (Bk (x[1], x[2], ..., x[k]))  
        {  
            if (x[1], x[2], ..., x[k] is a path to an answer node)  
                output x[1:k];  
            if (k < n) Backtrack(k+1);  
        }  
    }  
}
```